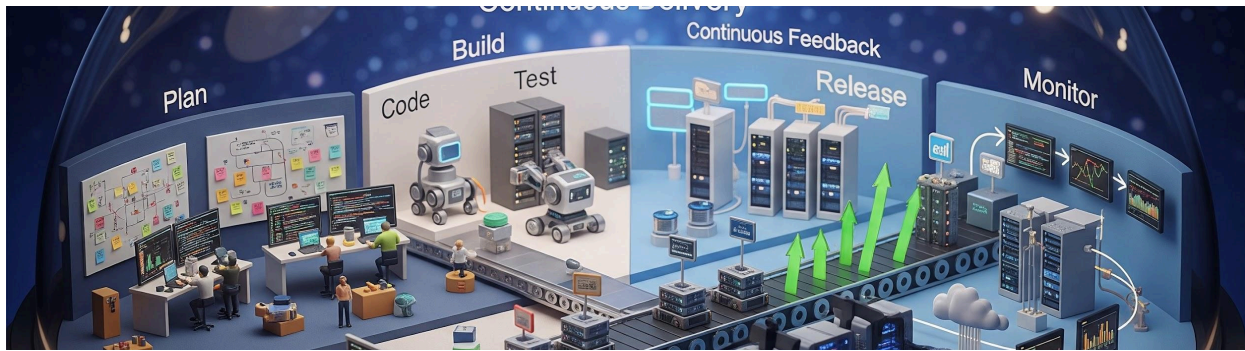




DevOps – Introduction

What is DevOps?

DevOps is a culture and set of practices that brings Development (Dev) and Operations (Ops) teams together to deliver software faster, reliably, and continuously.

Earlier, developers wrote code and operations teams deployed and maintained it. This separation caused delays, communication issues, and frequent failures. DevOps solves this problem by improving collaboration, automation, and continuous delivery.

In simple words: **DevOps = Development + Operations working together with automation.**

Traditional Model (Before DevOps)

In the traditional software development model:

1. Developers write code
2. Code is sent to operations team
3. Operations deploy it on servers
4. If something breaks → blame game starts

Problems in this model:

- Slow releases
- Miscommunication
- Manual deployment errors
- Lack of automation

The example flow below shows that each step takes a long time.

Step	Time Taken
Developer	Long time
Testing	Long time
Operations	Long time
Deployment	Long time

DevOps Model

DevOps improves this process by introducing automation and continuous integration.

In DevOps:

- Developers and operations teams collaborate
- Code is tested automatically
- Deployment is automated
- Monitoring happens continuously

DevOps Lifecycle

The example flow below shows the typical DevOps Lifecycle.

Stage	Activity
Plan	Requirement gathering
Code	Writing the application
Build	Compiling the application
Test	Automated testing
Deploy	Automated release
Monitor	Performance tracking

Benefits of DevOps

DevOps helps organizations:

- Deploy applications faster
- Reduce human errors
- Improve collaboration
- Increase system reliability
- Deliver continuous updates to users

Many companies like Amazon, Netflix, and Google use DevOps practices to release updates frequently.

DevOps Tools Overview

DevOps uses different tools for different stages of the lifecycle. A few examples of tools are listed below:

Stage	Tool
Code	Git, Github, Bit bucket
Build	Maven
CI/CD	Jenkins
Container	Docker
IAC-Infrastructure As a Code	(IAC) Terraform
Orchestration	Kubernetes
Monitoring	Prometheus, Grafana, Data dog

For this course we will focus mainly on:

- Git
- GitHub
- Jenkins - Maven
- Docker
- Kubernetes

- Prometheus
- Terraform
- Ansible

DevOps + AWS

AWS helps implement DevOps easily because it provides services like:

- EC2 for servers
- S3 for storage
- CodePipeline for CI/CD
- CloudWatch for monitoring

Simple Example

Without DevOps:

- Developers send code manually
- Operations deploy manually
- Errors happen often

With DevOps:

- Code pushed to GitHub
- Jenkins automatically builds it
- Docker runs the application
- Deployment happens automatically

One Line Summary: **DevOps helps teams deliver software faster through collaboration and automation.**